

Project: System Liquer Store

Team Name: LiquorTrack

Isaac Sebastian Acurio Catota

Demian Rolando Armas Guaño

Enrique Mateo Ayala Lagunas

University of the Armed Forces – ESPE

NRC: 36862

Object Oriented Programming

Instructor: Edison Lascano

October 7, 2026

Quito – Ecuador

Problem

Currently, the liquor store manages its business manually, without a system that centralizes information. Inventory is controlled by direct observation, so the owner often finds out too late that a product has run out and cannot place orders on time. Credit sales, which are common during events or large purchases, have no organized record, making it hard to know who owes money, how much they owe, and whether they have already paid. Promotions and discounts are applied by hand, which causes charging errors and confusion about when each one ends. Since the business is run by the owner and a single employee, it is difficult to know who made each sale and how much each person sold, and there is no reliable cash closing that allows comparing sales with the money available. On top of this, supplier information and pending orders are not organized, and there is no easy way to identify the best-selling products in order to decide what to buy. These shortcomings lead to financial losses, errors, and wasted time that hold back the growth of the business.

Overview

As a direct response to the manual limitations and disorganization we currently face, this project arose from the need to regain control over our business. The primary goal of the System Liquor Store is to digitize all our daily operations, replacing our flawed manual methods with efficient, centralized business management.

To solve our daily problems, the software will first provide a real-time view of our inventory to prevent stockouts of our best-selling bottles, while also automatically organizing supplier information and pending orders. Furthermore, it will feature a dedicated section for credit sales, allowing us to track exactly who owes money and who has already paid. To fix our cash closing issues, the system will distinguish between sales made by the owner and those made by the employee, ensuring an organized, reliable, and stress-free end-of-day reconciliation. Finally, by automatically applying promotions and discounts, we will eliminate the manual billing errors that have been affecting us. Ultimately, this system is the exact solution we need to overcome our current shortcomings, stop wasting time and money, and position our store for real growth.

Background

In a liquor store, daily operations require proper control of inventory, sales, customers, employees, suppliers, and payments. The sales process involves finding products quickly by name, category, or price, applying the correct promotions or discounts, verifying the customer's age, and recording who made each sale. At the same time, inventory must be monitored to identify products that are running low, while supplier information and pending purchase orders need to be organized to ensure that products can be restocked on time and the inventory remains updated after each delivery.

The business also requires an organized way to manage credit sales, record customer payments, and perform an accurate cash closing at the end of each shift. Daily, weekly, and monthly sales reports, together with information about the best-selling products, can help the owner monitor the performance of the business and make better purchasing and restocking decisions. Therefore, the proposed system will centralize these activities, making daily operations more organized, reducing manual errors, and providing reliable information for better control of the liquor store.